ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 7: Coordinate Geometry (Nirdeshank Jyamiti)

Key Concepts & Important Formulas:

* Distance Formula: Distance between P(x1, y1) and Q(x2, y2) is PQ = sqrt((x2 - x1)^2 + (y2 - y1)^2).

* Distance from Origin O(0, 0) to P(x, y) = sqrt(x^2 + y^2).

* Section Formula: Coordinates of point P(x, y) dividing AB in ratio m1:m2 internally:
x = (m1*x2 + m2*x1) / (m1 + m2), y = (m1*y2 + m2*y1) / (m1 + m2).

* Mid-point Formula: ((x1 + x2)/2, (y1 + y2)/2).

Solved Exam Examples & Practice:

Example: Find distance between (2, 3) and (4, 1).
d = sqrt((4 - 2)^2 + (1 - 3)^2) = sqrt(4 + 4) = sqrt(8) = 2*sqrt(2) units.